

The Role of AI in Sustainability and Climate Change

Artificial intelligence is a very controversial topic when it comes to sustainability. Its rapid growth requires increasingly powerful computing infrastructure, raising concerns about energy consumption, water use and electronic waste. However, focusing only on these costs overlooks the other side of AI's environmental impact.

The same technology that consumes significant resources can also be used to reduce waste, improve efficiency and help address climate change across some of the world's most resource-intensive sectors. So, the environmental impact of artificial intelligence depends not simply on how much energy it consumes, but on how it is developed, powered and applied.

One of the clearest examples of this is the management of renewable energy systems. Solar panels only produce electricity during the day, and wind turbines only generate power when the wind is blowing. Because renewable energy supplies naturally change, balancing electricity production and demand can be difficult. Artificial intelligence can analyze weather forecasts, cloud cover, wind speeds, battery storage levels and electricity usage patterns to optimize energy distribution. This reduces the need for fossil-fuel power stations that would otherwise have to compensate for changes in renewable generation.

The same idea to reduce unnecessary energy consumption can also be applied to smart buildings. Heating, ventilation, air conditioning and lighting systems often operate on fixed schedules regardless of whether anyone is in the building. Artificial intelligence-controlled systems can automatically adjust temperatures, airflow and lighting depending on how many people are inside, and outside weather conditions. This reduces electricity use while keeping the inside of the building comfortable.

Beyond energy systems and buildings, artificial intelligence can improve efficiency in manufacturing. Instead of waiting for machines to break, artificial intelligence can continuously monitor how equipment is operating and detect small even the smallest signs of wear and tear. Early maintenance prevents failures, reduces waste and lowers overall energy use. Once again, the environmental benefit comes from using information more efficiently to prevent unnecessary resource consumption.

Further, moving to transportation, artificial intelligence can optimize traffic flow through traffic signals, reducing congestion and vehicle idling times. Delivery companies can use AI to find shorter and more fuel-efficient routes and airlines can use intelligent systems to optimize flight paths and reduce unnecessary fuel consumption. Even small improvements become significant when they are applied across millions of journeys.

However, artificial intelligence does not only reduce emissions by making existing systems more efficient. It has also become a valuable tool for understanding the environmental problems themselves. Climate change involves complex interactions between the atmosphere, oceans, forests, ice sheets and human activities. Scientists collect huge amounts of data from satellites, weather stations, ocean sensors, aircraft and research expeditions, and analyzing all of this data manually would take an immense amount of time; and here, AI helps researchers process this information much faster. Improved climate models can then make more accurate predictions of future temperature changes, rainfall patterns, drought risks, flooding events and sea-level rise, much faster. And these predictions also have practical uses – more accurate forecasts can help governments prepare for natural disasters, while earlier warnings can reduce damage and protect human lives. Artificial intelligence can also help emergency services predict wildfire risks based on vegetation conditions, temperature, humidity and wind speed. Earlier detection and prevention of wildfires can limit environmental damage and prevent large amounts of carbon dioxide from entering the atmosphere.

When applied off-earth, artificial intelligence systems can analyze thousands of satellite images every day to identify deforestation, habitat destruction, coastal erosion and land-use changes. Forests absorb large amounts of carbon dioxide, so protecting them is essential for slowing climate change. Faster identification of illegal or damaging logging can allow governments to intervene before more extensive damage is done. This ability to monitor ecosystems also contributes to biodiversity conservation. Artificial intelligence can help identify endangered species, analyze migration patterns and monitor ecosystem health. Protecting biodiversity strengthens ecosystems that naturally absorb carbon and maintain environmental stability. In this way, AI can support not only the reduction of emissions but also the protection of natural systems that regulate the climate.

Agriculture is another sector where this combination of monitoring and optimization can reduce environmental impacts. Farming contributes significantly to greenhouse gas emissions, and at the same time, agriculture faces increasing challenges from rising temperatures, droughts, floods and changing weather patterns caused by climate change.

Artificial intelligence enables precision farming, in which farmers manage crops using detailed environmental information. Sensors can collect data about every part of the farms, and artificial intelligence can then analyze this data to determine where water, fertilisers or pesticides are actually needed. Using resources only where necessary can substantially reduce waste. Lower fertiliser use can decrease nitrous oxide emissions, a greenhouse gas far more powerful than carbon dioxide. Efficient irrigation conserves freshwater while also reducing the electricity needed to pump water across farms.

Healthcare provides another example of this principle, despite itself relying heavily on energy-intensive infrastructure. Hospitals consume large amounts of electricity through medical equipment, lighting, ventilation systems, refrigeration, laboratories and patient-care facilities. Artificial intelligence can improve efficiency across many of these services. It can assist doctors by analysing medical images such as X-rays, CT scans and MRIs rapidly. Earlier diagnoses can allow treatment before diseases become more severe, potentially reducing hospital admissions and repeated procedures. Artificial intelligence also supports drug discovery by analysing large biological datasets and helping researchers identify promising medicines more quickly.

Hospitals can also use it to optimize staffing schedules, equipment usage, patient admissions and operating-theatre availability. Improved efficiency can reduce wasted energy while improving patient care. Artificial intelligence can also predict equipment failures, extending the lifespan of medical devices and reducing unnecessary replacement and manufacturing.

Across all these fields, the same pattern emerges: artificial intelligence can reduce environmental impacts by using data to make complex systems more efficient. However, these benefits do not erase the environmental costs of artificial intelligence itself.

Artificial intelligence systems are being used increasingly across different industries, and as they become more powerful, they often require more computing capacity and more data for training and operation. This infrastructure consumes significant amounts of electricity. If the growth of artificial intelligence is not supported by a transition towards low-carbon and renewable energy sources, emissions from computing infrastructure could increase substantially.

Water consumption is another concern. Large data centres can require significant amounts of water for cooling, potentially placing additional pressure on water supplies in areas already facing scarcity. The location of these facilities therefore matters, as does the development of cooling technologies that use less water.

Electronic waste presents a further problem. Computers, specialized chips and other equipment can become outdated as artificial intelligence technology advances. Without effective systems for reusing and recycling this equipment, the rapid expansion of AI infrastructure could create additional environmental damage.

These concerns demonstrate why artificial intelligence is not automatically good for the environment. Its environmental impact depends on how governments, companies and researchers develop and use AI systems. The efficiency gains created by artificial intelligence can be weakened or even outweighed if the infrastructure supporting it is unnecessarily resource-intensive.

Artificial intelligence, therefore, should not be seen simply as either a problem or a solution for the environment. It is a tool whose environmental consequences depend on how it is built and

used. If artificial intelligence is powered primarily by fossil fuels, consumes resources inefficiently and develops without appropriate environmental safeguards, it could contribute further to climate change. If it is developed responsibly, powered increasingly by low-carbon energy and applied where it can produce meaningful efficiency gains, it could become a valuable tool for reducing emissions and addressing climate change.

Ultimately, the environmental impact of artificial intelligence will depend on whether the benefits it creates are matched by efforts to reduce the footprint of the infrastructure that makes those benefits possible. If AI systems are powered by cleaner energy, designed to use resources efficiently and governed by environmentally responsible policies, they can become an important part of reducing global emissions. Rather than treating artificial intelligence as inherently harmful or inherently sustainable, the focus should be on developing it into a tool for building a more sustainable future.